

Gender identification based on human brain structural MRI with a multi-layer 3D convolution extreme learning machine

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Abstract: Previous group-level neuroimaging studies have shown significant gender differences in the human brain. Research on sex-specific brain differences in healthy individuals is an important base for understanding sex-specific expression in psychiatric disorders. This study proposes a multi-layer 3D convolution extreme learning machine (MCN-ELM) to classify male and female brains based on structural MRI (sMRI) grey matter (GM) data scans from human connectome projects (HCP) of 876 healthy adults (491 females). First, the authors extracted multi-scale features by three-scale multi-layer 3D convolution neural networks (CNNs) without fine-tuning the parameters of convolution kernels. Then, they pulled the network output feature maps into a vector as separate ELMs. By voting on the three-scale networks, the MCN-ELM algorithm classifies male and female brains with an accuracy of 98.06% through a 10-fold cross-validation strategy, outperforming other state-of-the-art algorithms. The proposed method may be used to understand other brain diseases. Additionally, the results show that the human brain can be categorised into two distinct classes, male and female brains, suggesting it is better to treat men and women separately when researching psychiatric disorders.

1 Introduction

Gender plays an important role in clinical research, which can guide the results of scientific experiments [1]. In general studies, quantitative matching of gender-specific experiment data is used to eliminate the influence of gender. However, it is unclear whether such an approach works completely.

Research on sex-specific brain differences in healthy individuals is an important base for understanding sex-specific expression of psychiatric disorders. Previous studies have shown that gender differences exist in many behaviours and cognitive domains. Men usually outperform in visuospatial tasks [2, 3], while women tend to perform better in verbal fluency [4], emotional memory [5], and facial emotion recognition tasks [6]. Although there are many structural similarities in male and female brains, there are also various neuroanatomical features that differ between men and women [7, 8]. It has been demonstrated that most men have a larger brain size than women [9].

Additionally, grey and white matter (WM) volumes vary by gender. Most men have a lower percentage of grey matter and a higher percentage of WM and cerebrospinal fluid (CSF) than women [10]. However, based on several large datasets of MRI images, Joel *et al.* demonstrated extensive overlap between the distributions of females and males for many brain characteristics and concluded that human brains cannot be categorised into two distinct classes, male and female brains [11]. Their article is only analysed at the voxel level without considering whether all structural information is classified.

In recent years, there has been increasing interests in the use of machine-learning methods to analyse neuroimaging data [12–17]. In the field of gender classification research, Wang *et al.* combined principal component analysis (PCA) [18] and support vector machines (SVM) [19] to explore gender-related brain differences in a group of 140 healthy young individuals and achieved a mean classification accuracy of 89% by using both structural and functional MRI data [20]. Although the PCA method can extract some useful classification features from structural and functional MRI data, PCA has some limitations because it ignores the local structural information data [21]. Yuan *et al.* further proposed a novel three-dimensional feature descriptor to approach structural

brain image classification by combining local information on the signal intensity and the spatial relationship between features, resulting in better performance when identifying an individual's gender [22]. However, the computation is complex and consumes a large amount of computing resources.

Recently, deep learning has achieved success in many domains. In particular, deep convolution neural networks (CNNs) have been proven powerful for many computer vision tasks in which convolution operations play important roles [23]. Meanwhile, a few deep-learning algorithms have been developed in the field of medical imaging to solve the problem of disease diagnosis [24, 25] and data processing [26]. Although the deep-learning algorithm outperforms the traditional algorithm in many domains, there remain many restrictions. It is well known that deep-learning requires many samples to train the network. However, it is difficult to collect so many samples in some fields of brain imaging research. In addition, the deep-learning algorithm computational cost during both training and recognition processes is expensive [27]. Therefore, for many small sample data sets, deep learning has not outperformed other algorithms.

Extreme learning is a kind of single-layer feedforward neural network (SLFN). The weights between the input layer and hidden layer are randomly initiated and then fixed without iterative tuning [28]. The only parameters that need to be learned are the connection weights between the hidden layer and output layer. Therefore, a back-propagated (BP) algorithm [29] is not required to update the weight of the network layer by layer. As a result, ELM has less computational overhead than traditional neural networks. It is worth noting that ELM has a higher efficiency than other traditional machine-learning classification algorithms. Many studies have shown that ELM has better or considerable generalisation ability than SVM or its variants [30, 31].

In this paper, we proposed a multi-layer 3D convolution extreme learning machine (MCN-ELM) to classify HCP healthy brain sMRI grey matter (GM) data from 385 males and 491 females. First, we designed three-scale multi-layer 3D CNNs to extract multi-scale features. Different scales of networks have different size convolution kernels with parameters that are randomly generated and then fixed without tuning. Then, we pulled the network output feature maps into a vector as the input of the

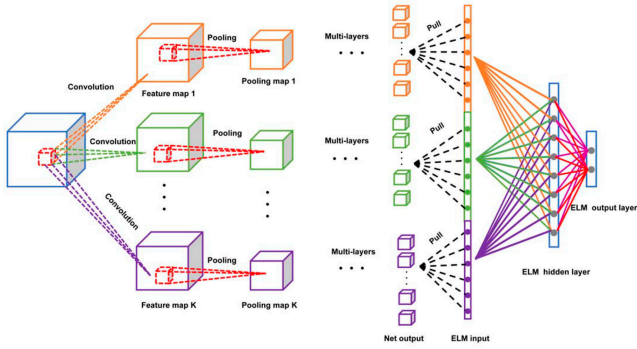


Fig. 1 Flowchart of the MCN-ELM algorithm. The original brain sMRI data are processed by multi-layer 3D convolution and 3D pooling operations. Then, the net output feature maps are pulled into a vector as the ELM input. Finally, there is an ELM classifier output label of the sample

ELM classifier separately. Finally, the final label of the sample was obtained by voting on three ELM classifier output labels. By efficiently and effectively discriminating males from females, our proposed method may provide an alternative approach for structural MRI research and offer a potential auxiliary diagnosis of brain disorders with structural changes.

2 Methods

Human brain sMRI consists of three-dimensional data with >2 million voxels, which is too large for general algorithms. Therefore, we propose an MCN-ELM algorithm to reduce dimension and extract effective structural features. Fig. 1 shows the algorithms flowchart. First, the original data are proposed by multi-layer 3D convolution and 3D pooling operations whose kernels are 3D, and the parameters are randomly initiated and then fixed without tuning. Then, we pull the network output feature maps into a vector as the input of the ELM. To make the results more believable, a 10-fold cross-validation strategy is used for training and testing data.

2.1 Subjects

The MRI data were obtained from human connectome projects (HCP, <http://www.humanconnectome.org/>). HCP is a major endeavour to map the neural pathways that underlie brain function and its variability in humans. The purpose of HCP is to acquire and share data about the structural and functional connectivity of the human brain, the connectome, on the macroscopic scale (~1 mm³ or greater) as analysed using non-invasive neuroimaging in a large population of healthy adults [32]. At present, the data include 876 healthy individuals: 385 males and 491 females.

2.2 Pre-processing

The data were processed by Statistical Parameter Mapping 8 (SPM8) toolbox and DARTEL package (Wellcome Department of Imaging Neuroscience, University College London, UK, <http://www.fil.ion.ucl.ac.uk/spm>). First, the skull for the structural MRI data was stripped. Then, the data were segmented into WM, CSF, and GM. Afterwards, the GM was normalised into Montreal Neurological Institute (MNI) space. Finally, to reduce the image noise, the GM data were smoothed by a Gaussian with a 3-mm full-width half-maximum kernel. This paper uses the GM data alone as the input data for the MCN-ELM algorithm.

2.3 Convolution

Convolution is an effective feature extraction method, which plays an important role in deep learning. Next, let us introduce the computational details of convolution operations. After convolution layer processing, each input 3D data (with resolution $d1 \times d2 \times d3$) generates n feature maps (with resolution $p1 \times p2 \times p3$). Specifically, the convolution kernel weight W is randomly generated, $W \in R^{c1 \times c2 \times c3 \times n}$. Each kernel size is $c1 \times c2 \times c3$, and

the weight of m th feature map is denoted as $w^m \in R^{c1 \times c2 \times c3}$ $m = 1, \dots, n$. For the input data, the node (i, j, k) value of m th feature map is computed by

$$l_{i,j,k,m} = \sum_{p=1}^{c1} \sum_{q=1}^{c2} \sum_{t=1}^{c3} x_{i+p+s1-1, j+q+s2-1, k+t+s3-1} w_{p,q,t}^m \quad (1)$$

$s1, s2$, and $s3$ in the formula denote the convolution kernel strides of three dimensions.

The feature map resolution is computed by

$$p1 = \frac{d1 - c1}{s1} + 1, p2 = \frac{d2 - c2}{s2} + 1, p3 = \frac{d3 - c3}{s3} + 1 \quad (2)$$

2.4 Pooling

The pooling operation can not only exponentially reduce the data dimension but also maintain the image rotation invariance. The invariance comes from the image rotating a tiny amount, and most output values of the pooling operation will not be changed. Local invariance is an important property, especially when we care about whether a feature is present instead of where it is. Denote $z1 \times z2 \times z3$ as the pooling kernel size, and the input data resolution is $p1 \times p2 \times p3$. Thus, the output pooling map size is $(p1/z1) \times (p2/z2) \times (p3/z3)$, and the node (α, β, γ) value of the m th pooling map is computed by

$$h_{\alpha,\beta,\gamma,m} = \text{Pooling}(l_{i,j,l,m}) \quad (3)$$

$$\begin{aligned} i &\in [(\alpha - 1) \times z1, \alpha \times z1], j \in [(\beta - 1) \times z2, \beta \times z2], \\ l &\in [(\gamma - 1) \times z3, \gamma \times z3] \end{aligned} \quad (4)$$

The pooling operations mainly contain max pooling and average pooling. This paper chooses average pooling for the network.

2.5 Extreme learning machines

ELM was proposed by Huang *et al.* [33]. They consist of a SLFNs where the weights between the input and hidden layer are randomly generated as long as the neuron activation functions are non-linear piecewise continuous, and they are then fixed without tuning. The only parameters that need to be calculated are the weights, which are between the hidden layer and output layer. There are two steps for using ELM to solve a problem: (i) feature mapping and (ii) parameter solving. In the first step for the ELM map, the data are input into a feature space by randomly initialising the weights between the input data and hidden layer. For the given input data, the output function of ELM is

$$f_L(x) = \sum_{i=1}^L \beta_i h_i(x) = h(x) \beta \quad (5)$$

where $\beta = [\beta_1, \dots, \beta_L]^T$ representing the weights between the hidden layer of L nodes and the output m nodes. $h(x) = [h_1(x), \dots, h_L(x)]$ is the output feature of the hidden layer. In fact, $h_i(x)$ is any non-linear piecewise continuous neurons activation function, and it can be described as

$$h_i(x) = G(a_i, b_i, x), a_i \in R^d, b_i \in R \quad (6)$$

where G is the activation function of the ELM, which helps to optimise the whole neural network, and in this study, it is set to be a simple sigmoid function. In the second step, the weights β between the hidden layer and output are solved by minimising the squared error. T indicates the labels of the input data.

$$\min_{\beta \in R^{L \times m}} \| H\beta - T \|^2 \quad (7)$$

By calculating $\beta^* = H^+ T$, we got the solution of the aforementioned equation, where H^+ denotes the Moore–Penrose generalised inverse of the matrix H , and

$$T = \begin{bmatrix} t_1^T \\ \vdots \\ t_N^T \end{bmatrix} = \begin{bmatrix} t_{11} & \cdot & \cdot & \cdot & t_{1m} \\ \vdots & & & & \vdots \\ t_{N1} & \cdot & \cdot & \cdot & t_{Nm} \end{bmatrix}_{N \times m} \quad \text{and} \quad H = \begin{bmatrix} h(x_1) \\ \vdots \\ h(x_N) \end{bmatrix}_{N \times L} \quad (8)$$

$$= \begin{bmatrix} h_1(x) & \cdot & \cdot & \cdot & h_L(x) \\ \vdots & & & & \vdots \\ h_1(x_N) & \cdot & \cdot & \cdot & h_L(x_N) \end{bmatrix}_{N \times L}$$

2.6 Network architecture

To obtain more useful features, we design three different scales multi-layer 3D convolution networks to separately extract multi-scale features. Then, we separately pull the output feature maps of networks into a vector as the input of ELM. Finally, the labels of brain image data are voted by the result of three different networks. The multi-layer convolution (MCN) is similar to an operation of dimension reduction. To highlight the effectiveness of MCN operation, this paper provides two-dimension reduction methods, PCA and paired *t*-test, for comparison. Additionally, the ELM is compared with the SVM, k-nearest neighbour (KNN) algorithm [34], and linear discriminant analysis (LDA) algorithm [35]. The average accuracy is separately calculated by 10-fold cross strategies. We obtain the result using MATLAB 2013b, which runs on a computer with an Intel(R) Xeon(R) CPU E5-2650 v3 @2.30 GHz.

Tables 1–3 show the details of the three different scale MCN networks. Rectified linear units are used for all nodes. Table 1 shows the parameters of the small-scale network. The size of the convolution kernels is comparably small, which is designed to extract detailed information. Table 2 shows the detail information

of a mid-scale network that is used to extract local information. Table 3 shows parameters of the large-scale network that is used to extract global information. The output feature map sizes of three networks are $12 \times 10 \times 12 \times 13$, $10 \times 12 \times 10 \times 15$, and $10 \times 8 \times 10 \times 20$. Then, we pull the network output feature maps into a vector as the ELM input data. By combining the two algorithms, we got both advantages of feature extracting and discriminating while controlling the consumption of time and computer source.

3 Results

To facilitate the presentation of convolution feature maps with different scale convolution kernel sizes, Fig. 2 shows the feature maps of 2D convolution in which the kernel parameters are randomly generated and then fixed without tuning. From Fig. 2, we can see that different sizes of convolution kernels extract different scales features. With the increase in the convolution kernel, the output feature maps become increasingly abstracted.

3.1 Classification accuracy

The number of ELM hidden nodes plays an important role in the classification problem. Fig. 3 shows the ELM-based classification accuracy of three-scale MCN network output features. With increases in the hidden layer nodes, the classification accuracy of three networks and vote accuracy increases gradually, which tends to be stabilised with slight fluctuation. When the number of hidden nodes is >9000, the voting accuracy is above 97%. When the number of hidden nodes is 18,000, the MCN-ELM method has the highest voting accuracy of 98.06%.

Table 4 shows the accuracy of the KNN, LDA, SVM, and ELM algorithm on three-scale network (MCN) output features. It can be seen that the small-scale network output features have the highest discrimination, and mid-scale outperforms large scale. It can also be seen that no matter what kind of classification algorithm, the final voting accuracy is higher than three-scale accuracy.

Table 1 Parameters of the small-scale MCN network

Layers	Input size	Kernel size	Maps	Stride	Out size
1 Convolution	$121 \times 145 \times 121 \times 1$	$4 \times 6 \times 4$	50	1	$118 \times 140 \times 118 \times 50$
2 Pooling	$118 \times 140 \times 118 \times 50$	$2 \times 2 \times 2$	50	2	$59 \times 70 \times 59 \times 50$
3 Convolution	$59 \times 70 \times 59 \times 50$	$4 \times 5 \times 4$	40	1	$56 \times 66 \times 56 \times 40$
4 Pooling	$56 \times 66 \times 56 \times 40$	$2 \times 2 \times 2$	40	2	$28 \times 33 \times 28 \times 40$
5 Convolution	$28 \times 33 \times 28 \times 40$	$5 \times 4 \times 5$	13	1	$24 \times 30 \times 24 \times 13$
6 Pooling	$24 \times 30 \times 24 \times 13$	$2 \times 3 \times 2$	13	$2 \times 3 \times 2$	$12 \times 10 \times 12 \times 13$

Table 2 Parameters of the mid-scale MCN network

Layers	Input size	Kernel size	Maps	Stride	Out size
1 Convolution	$121 \times 145 \times 121 \times 1$	$10 \times 12 \times 10$	55	1	$112 \times 134 \times 112 \times 55$
2 Pooling	$112 \times 134 \times 112 \times 55$	$2 \times 2 \times 2$	55	2	$56 \times 67 \times 56 \times 55$
3 Convolution	$56 \times 67 \times 56 \times 55$	$13 \times 14 \times 13$	40	1	$44 \times 54 \times 44 \times 40$
4 Pooling	$44 \times 54 \times 44 \times 40$	$2 \times 2 \times 2$	40	2	$22 \times 27 \times 22 \times 40$
5 Convolution	$22 \times 27 \times 22 \times 40$	$13 \times 16 \times 13$	15	1	$10 \times 12 \times 10 \times 15$

Table 3 Parameters of the large-scale MCN network

Layers	Input size	Kernel size	Maps	Stride	Out size
1 Convolution	$121 \times 145 \times 121 \times 1$	$32 \times 38 \times 32$	55	1	$90 \times 108 \times 90 \times 55$
2 Pooling	$90 \times 108 \times 90 \times 55$	$2 \times 2 \times 2$	55	2	$45 \times 54 \times 45 \times 55$
3 Convolution	$45 \times 54 \times 45 \times 55$	$36 \times 47 \times 36$	15	1	$10 \times 8 \times 10 \times 20$

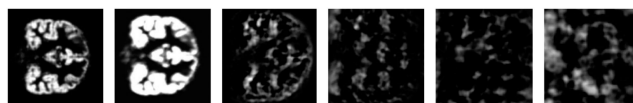


Fig. 2 Feature maps of different scales convolution kernels on the original data. From left to right: the original data with size of 121×121 , the feature map convolved with the 5×5 kernel, the feature map convolved with the 15×15 kernel, the feature map convolved with the 25×25 kernel, the feature map convolved with the 35×35 kernel, and the feature map convolved with the 45×45 kernel

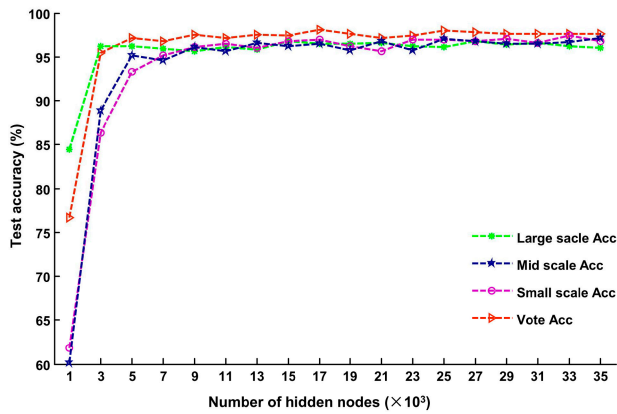


Fig. 3 ELM-based classification accuracy of three scale network output feature maps with increasing hidden layer nodes

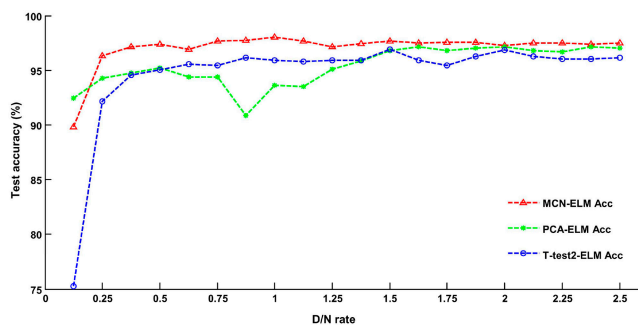


Fig. 4 Accuracy of feature extraction algorithms MCN, PCA, and paired t -test (t -test2) with an increasing N/D rate

For ELM-based classification, to compare the effectiveness of different dimension reduction methods, this paper proposes the N/D ratio. N is the number of hidden nodes of ELM. D is the data dimension of the input ELM classifier. Fig. 4 shows the accuracy with increasing the N/D ratio. For the paired t -test method, we set $p = 0.000001$ and then select the top 12,000 as the ELM input data. For the PCA method, the dimension of input ELM is ~ 788 . To facilitate the calculation, we approximate the input data dimension D as 800. For the MCN method, three-scale dimensions are 18,720, 18,000, and 16,000 separately, and we approximate the input data dimension D as 16,000. By increasing the N/D ratio, the accuracy increases and tends to be stable. The max accuracy of MCN is 98.06%. Additionally, the max accuracy of paired t -test is 96.78%. The max accuracy of PCA is 97.14%. The result shows that the MCN method is more effective than PCA and paired t -test.

Table 5 shows the given classification algorithm performance for paired t -test, PCA, and the MCN feature extraction method. The following conclusions can be drawn from Table 5. The ELM algorithm achieves the highest recognition accuracy for each type of dimension reduction method. Additionally, the KNN algorithm has the fastest training speed and the LDA algorithm has the fastest recognition speed. The ELM algorithm with the MCN combination achieves the highest accuracy for gender classification.

4 Discussion

Previous group-level neuroimaging studies have shown significant gender differences in the human brain [6, 36]. In this paper, we proposed an MCN-ELM algorithm to identify male and female brains based on sMRI data. We designed three-scale MCN networks with different size convolution kernels and pooling kernels to extract multi-scale information. To make the result more convincing, a 10-fold cross-validation strategy is used during the calculation. By voting using three scales in the MCN-ELM network, the final voting accuracy is 98.06%. The result shows that the proposed method extracts sufficient information from sMRI data and reliable differences exist in male and female brains at the individual level.

Table 4 Accuracy of KNN, LDA, SVM, and ELM on three scale network output features

	Small-scale Acc., %	Mid-scale Acc., %	Large-scale Acc., %	Vote Acc., %
MCN-KNN	79.21 $\pm$ 2.90	76.91 $\pm$ 2.10	76.59 $\pm$ 4.25	81.47 $\pm$ 2.42
MCN-LDA	90.29 $\pm$ 3.45	84.11 $\pm$ 4.06	81.50 $\pm$ 5.40	91.76 $\pm$ 3.04
MCN-SVM	96.57 $\pm$ 1.53	95.55 $\pm$ 1.34	94.56 $\pm$ 2.33	96.91 $\pm$ 1.34
MCN-ELM	96.94 $\pm$ 1.42	96.45 $\pm$ 1.76	96.40 $\pm$ 1.75	98.06 $\pm$ 0.94

Table 5 Accuracy of different feature extraction algorithms on competing classifiers

Method	Feature extraction	Accuracy, %	Training time, s	Recognition time, ms
LDA-based	Paired t -test	81.29 $\pm$ 4.56	1880.9	9.84
	PCA	93.72 $\pm$ 2.37	3.140	1.77
	MCN	91.76 $\pm$ 3.04	9776	74.5
KNN-based	Paired t -test	87.29 $\pm$ 3.53	0.060	1888
	PCA	80.83 $\pm$ 3.42	0.016	105.4
	MCN	81.47 $\pm$ 2.42	0.319	5700
SVM-based	Paired t -test	96.68 $\pm$ 1.76	0.872	839.2
	PCA	96.68 $\pm$ 1.76	0.537	112.1
	MCN	96.91 $\pm$ 1.34	3.500	3000
ELM-based	Paired t -test	96.78 $\pm$ 1.18	10.30	280
	PCA	97.14 $\pm$ 2.26	0.429	27
	MCN	98.06 $\pm$ 0.94	34.00	1300

To facilitate the presentation of convolution feature maps, Fig. 2 shows the feature maps of different scales of 2D convolution in which the kernel parameters are randomly generated and then fixed without tuning. To extract more different features and match the data size, we set the convolution kernel size gap between different scale networks at ~ 10 . It can be seen that with the increase in the convolution kernel size, the feature maps become increasingly abstract. When the convolution kernel size is < 10 , the outline of the brain feature maps can be clearly seen. With increases in the convolution kernel size, the outline of the brain becomes blurred. When the convolution kernel size is above 20, the feature maps are too abstract to see the brain outline. Additionally, previous experiments have shown that the classification accuracy decreases with increasing the convolution kernel size. Therefore, we designed three scales of networks in the experiment. The final voting accuracy of the three networks outperformed the traditional method. When applying this proposed MCN-ELM method to solve other domain problems, more scale networks can be designed.

Based on the results, the differences in detailed structures between males and females are more obvious than local and global structures. Additionally, the features extracted from different scales of networks have a certain degree of independence. The kernel size of a small-scale network is smaller, and the convolution and pooling mainly focus on the smaller area of the brain structure. Therefore, we can suppose that the small-scale network mainly extracts detailed feature information. Similarly, the mid-scale network extracts local feature information, and the large-scale network extracts global feature information. It can be seen from Table 4 that no matter what kind of classification algorithm we use, the accuracy of the small-scale network is the highest, and the mid-scale network outperforms the large-scale network. Therefore, we can suppose that males and females have more obvious differences in the detailed structure. From Table 4, we can also see that the final voting accuracy is higher than the accuracy of the other three scales irrespective of the proposed classification method. This result confirms our conjecture that different scales of networks extract different feature information, and the features extracted

from the three-scale networks have a certain degree of independence.

Additionally, the multi-layer convolution network (MCN) is an effective dimension reduction and feature extraction method. To highlight the effectiveness of MCN, the PCA and paired *t*-test algorithms were compared with MCN. From Fig. 4, it can be seen that with increasing the N/D ratio, the accuracy increases and tends to be stable. Additionally, MCN is more effective than PCA and paired *t*-test. MCN may be an effective feature extraction method for evaluating 3D data. From Table 5, it can be seen that for SVM-based evaluation, the accuracies of the MCN, PCA, and paired *t*-test methods are equivalent. Additionally, for ELM-based evaluation, the MCN achieved the highest accuracy. However, for KNN and LDA, the MCN method does not achieve the highest accuracy. The combination of ELM and MCN achieves the highest accuracy among these combinations, as shown in Table 5. Different classification algorithms may be suitable to address different features and may be due to both the weights between the input and hidden layer of ELM and the parameters of MCN are randomly generated. The combination of ELM and MCN may be the most appropriate and achieve the highest discrimination.

It is well known that deep learning achieves great success in many domains, such as speech recognition, object detection, visual object recognition, genomics, and others. The use of big data is a prerequisite for deep learning. Deep learning requires many samples for iterative training to update the weights of the net [23]. In many research fields, especially in the medical domain, deep learning faces great challenges. First, funding for construction data sets is difficult to obtain. Second, the annotation of medical images requires expensive medical expertise. Third, privacy issues make it difficult to collect and share medical data compared to natural images. Fourth, the data collected by different data collection centres are slightly different and cannot be simply integrated [37]. Therefore, it is difficult to collect big data for medical images. As a result, the deep learning method is restricted in the medical image field. Some transfer learning methods were used for medical images when there are limited data [38]. However, it is difficult to find proper transfer data. The MCN-ELM network is similar to deep learning. The only difference between MCN-ELM and deep learning is that the weights of the MCN network are generated randomly and without tuning. The parameters that need to be calculated are the ELM weights between the hidden layer and output layer. Therefore, the computational cost is greatly reduced. Additionally, there is no need to collect big data for training to update the weights of the net. The MCN-ELM algorithm may be a successful approach when there are small sample data problems.

From the view of classification, this paper demonstrated that human brains can be categorised into two distinct classes: male and female brains. In studies on brain sMRI domains, Zhao *et al.* used a multi-layer vote algorithm and demonstrated that most slices of the brain can be classified and that the structural differences between males and females exist in the whole brain [39]. In studies using diffusion tensor imaging (DTI) domains, the diffusion-based structural connectome of the brain has novel sex differences. Females have greater between-hemispheric and cross-module connectivity, while within-hemispheric connectivity, as well as modularity and transitivity participation are more prominent in males. Additionally, in the studies of fMRI domains, many studies have shown that males and females have specific differences in brain resting-state fMRI data [20, 40]. Additionally, the age of onset, prevalence, and symptomatology of many neuropsychiatric conditions differ between males and females [41]. These results show that sex is an important factor and suggest that research and treatment of psychiatric disorders in males and females are ideally performed separately for each gender.

Considering the MCN-ELM algorithm, the architecture parameters of the network can be adjusted, including the number of the MCN layers and the size and number of convolution and pooling kernels. Several different settings were evaluated in the preliminary experiments. Additionally, the accuracy of different architecture networks has slight fluctuation. Future work on networks could focus on additional optimisation of these features. Furthermore, the number of hidden nodes plays an important role

in ELM. Evaluation of the many hidden layer nodes matching the data dimension is a worthwhile endeavour. Additionally, the experimental results show that different randomly initialised ELM weights have a slight effect on the final accuracy. Future work on ELM could focus on the approach for initialising better weights and evaluating the number of hidden nodes.

5 Conclusion

In conclusion, this paper proposed a MCN-ELM to classify healthy male and female brain structural MRI GM data. ELM outperforms most of state-of-the-art classification methods, including KNN, LDA, and SVM. Compared with PCA and paired *t*-test, the MCN is an effective feature extraction method. The combination of ELM and MCN achieves the highest classification accuracy of HCP brain MRI data. Our results show that the human brain can be categorised into two distinct classes: male and female brains. Additionally, gender is an important factor, suggesting that the research or treatment of psychiatric disorders in males and females would ideally be performed separately for each gender.

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